

PROJECTIONS OF LINES

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[1] A line AB 80mm long has its end A 20mm above HP and 30mm in front of VP. It is inclined at 30° to HP and 45° to VP. Draw the projections of the line and find apparent lengths and apparent inclinations.

[2] A line AB 80mm long is inclined to HP at 30° and inclined to VP at 45° . Draw the front and top views of a line and determine their lengths. Also, measure the perpendicular distance of end B from both HP and VP.

[3] Draw the projections of a line AB 100mm long inclined at 45° to VP and 30° to HP. One end of the line is 20mm above HP and in VP. Determine apparent lengths and inclinations.

[4] A straight line PQ 65mm long is inclined at 45° to HP and 30° to VP. The point P is 70mm from both the reference planes and point Q is towards the reference planes. Draw the projections.

[5] A line AB measuring 70mm has its end A 15mm in front of VP and 20mm above HP and the other end B 60mm in front of VP and 50mm above HP. Draw the projections of the line and find the inclinations of the line with the both the reference planes of projections.

[6] A line AB has its end A 20mm above HP and 15mm in front of VP. The other end B is 60mm above HP and 45mm in front of VP. The distance between end projectors is 70mm. draw its projections. Determine the apparent lengths and true inclinations.

[7] The end A of a line AB is on HP and 25mm in front of VP. The end B is on VP and 50mm above HP. The distance between the end projectors when measured parallel to the line of intersection of HP and VP is 65mm. Draw the projections of the line AB and determine its true length and true inclinations with HP and VP.

[8] The end A of a line AB is in HP and 25mm in front of VP. The end B is 10mm in front of VP and 50mm above HP. The distance between the end projectors when measured parallel to the line of intersection of HP and VP is 80mm. Draw the projections of the line AB and determine its true length and true inclinations with HP and VP.

[9] A line PQ 85mm long has its end P 10mm above HP and 15mm in front of VP. The top view and front view of line PQ is 75mm and 80mm respectively. Draw its projections. Also, determine the true and apparent inclinations of the line.

[10] The top view of a line AB, 80mm long measures 65mm and the length of the front view is 50mm. the end A is in HP and 15mm in front of VP. Draw its projections.

[11] The top view of a line PQ 75mm long measures 50mm and the front view measures 60mm. the end P is 30mm above HP and 15mm in front of VP. Draw the projections of the line and find its true inclinations with HP and VP. Find the distance between the end projectors.

[12] A line has its end A 10mm above HP and 15mm in front of VP. The end B is 55mm above HP and line is inclined at 30° to HP. The distance between the end projectors is 50mm. draw the projections of the line. Determine the true length of the line and its inclination with VP.

[13] Draw the projections of a line PQ and find its apparent length, true length and true inclinations with HP. When the line PQ has its end P 25mm above HP and 20mm in front of VP. The distance between the end projectors of the line when measured parallel to the line of intersection of the HP and VP is 60mm. the end Q is 50mm above HP and the line is inclined at 30° to the VP.

[14] The line PQ measures 80mm in length. The point P is above HP and in front of VP by 20mm and 30mm respectively. The distance between the end projectors is 50mm. the line is inclined to VP by 30° . Draw the projections of the line and specify its true inclination with HP.

[15] The top view of a line PQ is 70mm and front view is 60mm. the end Q is near to both HP and VP than the end P and is 15mm above HP and 20mm in front of VP. Draw the projections of the line if the distance between projectors is 50mm.

[16] The top view of a line PQ 75mm long measures 50mm. the end P is 30mm in front of VP and 15mm above HP. The end Q is 15mm in front of VP and above HP. Draw the projections of the line and find its inclinations with HP and VP. **Find the length of front view and distance between the end projectors.**

[17] The line has one end 30mm in front of VP and 15mm above HP and the other end is 15mm in front of VP and is above HP. Length of the line is 60mm. top view of the line is 40mm long. Draw the two views of the line and obtain the inclination of the line with HP and VP.

[18] The point B of a line AB is on horizontal plane, the top view of the line makes an angle of 30° with XY line, being 80mm. the point A is on the vertical plane and 50mm above the horizontal plane. Draw the top and front views of the line and obtain the true length of the line. Also, find the inclinations of the line with the two planes.

[19] The top view of a straight line AB is 60mm long and makes an angle of 30° with the XY line. The end A is in VP and 30mm above HP. The end B is 65mm above HP. Draw the projections of the line AB and determine (i) length of the front view (ii) its true length and true inclinations with the reference planes.

[20] The top view of PQ of a straight line is 70mm and makes an angle of 60° with XY line. The end Q is 10mm in front of VP and 30mm above HP. The difference between the distances of P and Q above HP is 45mm. draw the projections; determine its true length and true inclinations with HP and VP.

[21] A point P is 40mm above HP and 20mm in front of VP, another point Q is 20mm above HP and 50mm in front of VP. The top view of line PQ is inclined at 30° to XY. Draw the projections.

[22] A straight line AB is inclined at 45° to HP and 30° to VP. The point A is in HP and 35 mm in front VP and the point B is in VP. The length of the straight line is 65mm. draw the projections of the straight line AB.

[23] One end of a line is 30mm in front of VP and 30mm above HP. The line is inclined at 40° to HP and its top view measuring 60mm, is inclined at 50° to XY. Draw the projections of the line and determine true length and inclination with VP.

[24] The elevation (front view) of a line AB 90mm long is inclined at 30° to HP and measures 70mm, the end A is 20mm above HP and is in VP. Draw the projections of the line and find the inclination with VP.

[25] Find the true length and true inclination of a line AB with HP having one of its ends 20mm in front of VP and 30mm above HP. The line is inclined at 40° to VP and the right side view of the line is 60mm long and inclined at 60° to the X1 Y1 line. Draw all the three views of the line.

[26] A straight line PQ 80mm long appears to a length of 50mm and inclined at 30° to XY line in its profile view. Draw its projections when its end point P is 15mm above HP and 60mm in front of VP. Point Q is nearer to VP than P.